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Working Group III - Common ICs: Cosmological

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1. Working Group Description

1. Task

Provide a set of common initial conditions for cosmological zoom-in simulations, write a research article introducing such initial conditions

2. Targeted Goals

- We will provide initial conditions for high-resolution cosmological zoom-in simulations targeted at eight different types of halos. Four different halo masses: $\sim 10^{10}$, $\sim 10^{11}$, $\sim 10^{12}$, $\sim 10^{13}$ M_{sun} at $z=0$, and two different types of merger histories per each mass: violent (i.e. multiple major mergers) and quiescent (i.e. few major mergers). Common initial conditions will be generated with MUSIC which supports most of the participating codes.
- A strong isolation criterion will be imposed for the quiescent set of initial conditions to select target halos. That is, $3 R_{\text{vir}}$ radius circle of the halo being simulated must not intersect $3 R_{\text{vir}}$ radius circle of any halo with half or more of its mass at redshift 0. A relaxed condition for the violent set of initial conditions: $2 R_{\text{vir}}$ circle instead of $3 R_{\text{vir}}$, now that MUSIC has the masking support, see 3-C of [this page](#) (flagship DM-only comparison).

2. Pipeline to Generate Cosmological ICs

1. Philosophy Proposed

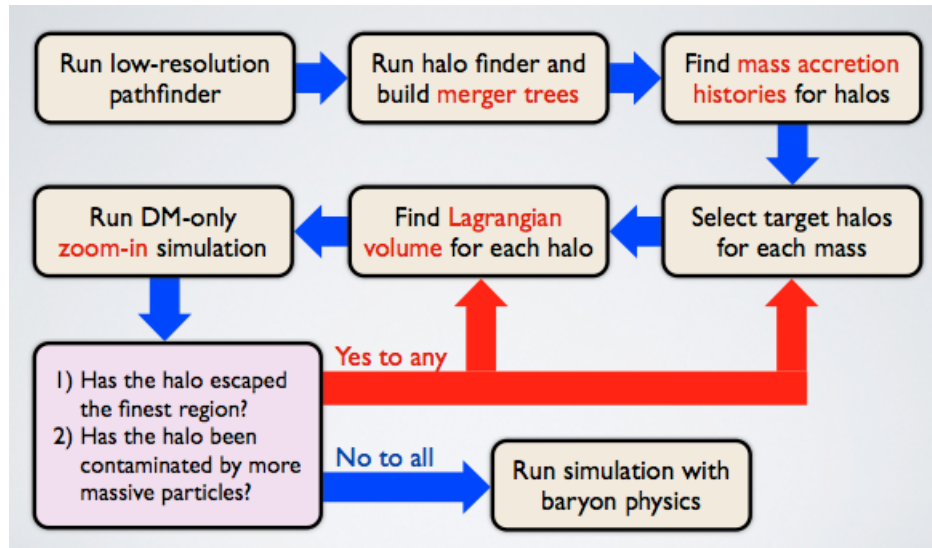
- Formation of dwarf-sized to elliptical galaxies and to galaxy groups in the high-resolution (< 100 pc physical) cosmological simulations including star formation and feedback
- Four types of halo masses

(e.g. maximum circular velocity $v_{\text{vir}} \sim 30 \text{ km/s}$ or $\sim 10^{10} M_{\text{sun}}$,
 $v_{\text{vir}} \sim 90 \text{ km/s}$ or $\sim 10^{11} M_{\text{sun}}$,
 $v_{\text{vir}} \sim 160 \text{ km/s}$ or $\sim 10^{12} M_{\text{sun}}$,
 $v_{\text{vir}} \sim 250 \text{ km/s}$ or $\sim 10^{13} M_{\text{sun}}$ @ $z=0$)

– Two types of merger histories

(e.g. one with a relatively violent merger history, another with fewer mergers)

2. Pipeline: Steps to Generate ICs



3. Adopted Cosmology

– LambdaCDM based on WMAP7+BAO+H0 (Komatsu et al. 2011) :

$h=0.702$, $\Omega_M=0.272$, $\Omega_{\Lambda}=0.728$, $\Omega_b=0.0455$, $n_s=0.961$ and $\sigma_8=0.807$

other density components assumed to be zero: i.e. $\Omega_R=\Omega_k=0$

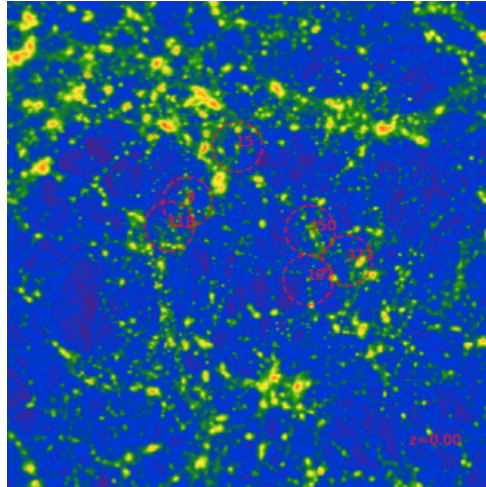
– An IC for a $\sim 1e11 M_{\odot}$ halo based on the Planck cosmology is also being tested (see [last Section](#) of this doc).

4. Low-resolution Pathfinder

– This description is for the $10^{11} - 10^{13} M_{\text{sun}}$ halos. For $10^{10} M_{\text{sun}}$ halos see the description in [WG VI](#).

– A $(60 \text{ Mpc/h})^3$ box created by MUSIC (Multi-Scale Initial Conditions generator, Hahn & Abel 2011)

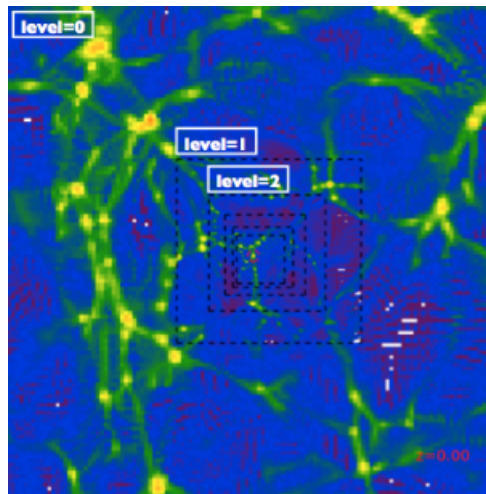
– 512^3 effective resolution in central $(42 \text{ Mpc/h})^3$ box, gravity resolution of 7 kpc/h



5. High-resolution Test

- This description is for the $10^{11} - 10^{13} M_{\text{sun}}$ halos. For $10^{10} M_{\text{sun}}$ halos see the description in [WG VI](#).
- Nested IC for each of the six target halos selected in low-resolution pathfinder
- Expediated generation of ICs with MUSIC !
- 128^3 top-grid resolution in a $(60 \text{ Mpc/h})^3$ box re-centered on a target halo
- IC with nested grids to impose high-resolution only around the target halo
- 4096^3 effective resolution in the central $(3 - 6 \text{ Mpc/h})^3$ box with $2.37 \times 10^5 M_{\text{sun}}/\text{h}$ particle resolution
- The finest resolution volume selected so that the target halo is not contaminated by any low resolution particles at the final redshift ($z=0$ for $\sim 10^{10} - 10^{12} M_{\text{sun}}$ halo, or $z=2$ for $\sim 10^{13} M_{\text{sun}}$ halo). However, the finest resolution volume should to as small as possible in order to minimize the computational cost.
- Note that MUSIC now has the masking support, see 3-C of [this page](#) (flagship DM-only comparison).

In the later sections, we present the dark matter only test of our ICs with Enzo or Gadget-2.



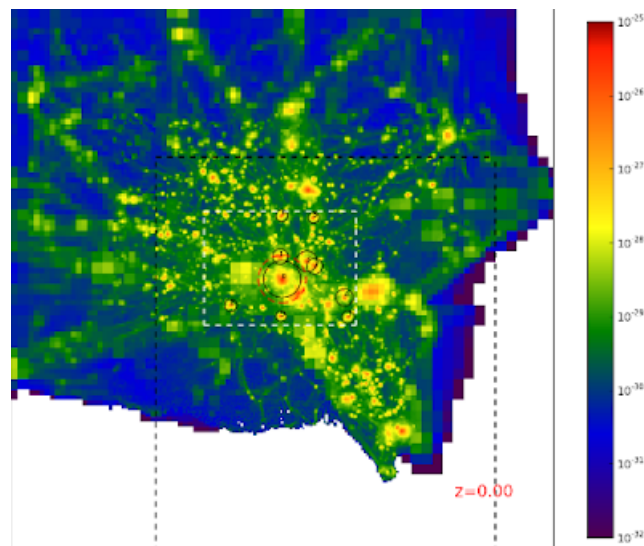
3. Chosen Halos: $\sim 10^{10}$ Ms Halos

- See the detailed description in [WG VI](#). You may find the MUSIC parameter files posted there as they are ready.

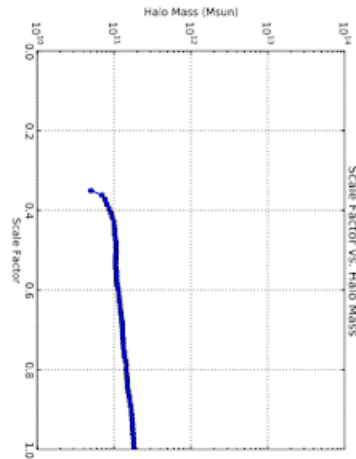
4. Chosen Halos: $\sim 10^{11}$ Ms Halos

[4-1] Quiescent Halo (Halo with A Quiescent Merger History)

- $M_{\text{vir}} \sim 1.8 \times 10^{11} M_{\text{sun}}$ @ $z=0$
(M_{vir} defined as the mass enclosed in a sphere of mean density contrast of 360 w.r.t. the background density, measured in the low-resolution simulation)
- Lagrangian region of a sphere of radius 375 kpc @ $z=0$
+ a buffer to prohibit contaminations by massive particles inside the halo @ $z=0$ ($256 \times 320 \times 256$)
- Density-weighted z-projection of DM density in a 6 Mpc/h box, now with finest resolution particles only
(black box: finest level at the start, white box: contains only the finest particles at the end of simulation, black circle: maximum extent of the particles of the target halo)



- Accretion history: main branch evolution of M_{200}
(M_{200} defined as the mass enclosed in a sphere of mean density contrast of 200 w.r.t. the critical density, measured in the low-resolution simulation)



– Movies of DM-only zoom-in simulation: z-projection of DM density in a 9 Mpc/h box

<http://www.jihoonkim.org/agora/z-movie-1e11q.mov>

– MUSIC parameter file (note it is for DM-only sims)

(1) [ics_60Mpc_1e11q_nocv_enzo.conf](#)

(NOTE: this is an old parameter with a cuboidal highest resolution region for our **Flagship Paper** by Ji-hoon Kim; not to be used any longer, still here only for an archival purpose; Nov. 20, 2012)

(2) [ics_AGORA_L12_1e11q_500kpc.conf](#)

(NOTE: this is a newly selected parameter for **Paper "DM-Highres"** by Oliver Hahn, of the same halo as above but with an ellipsoidal highest resolution region; change levelmax from 12 to 13 to increase resolution; Jan. 17, 2014)

[4-2] Violent Halo (Halo with A Violent Merger History)

– $M_{\text{vir}} \sim 3.2 \times 10^{11} M_{\text{sun}}$ @z=0

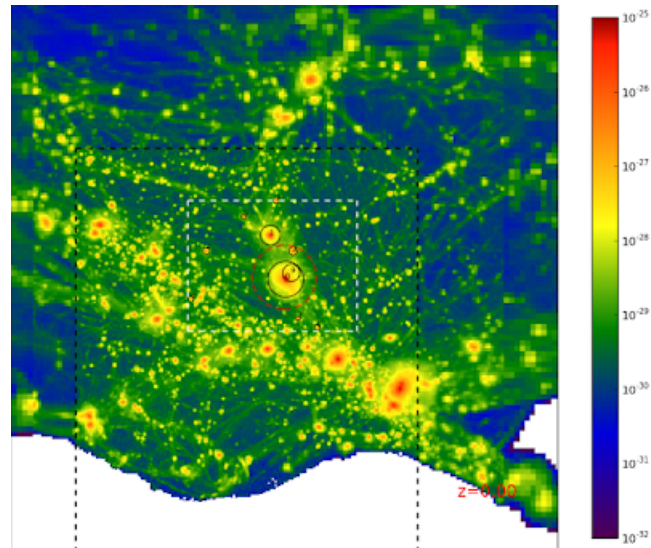
(M_{vir} defined as the mass enclosed in a sphere of mean density contrast of 360 w.r.t. the background density, measured in the low-resolution simulation)

– Lagrangian region of a sphere of radius 499 kpc @z=0

+ a buffer to prohibit contaminations by massive particles inside the halo @z=0 (256×320×448)

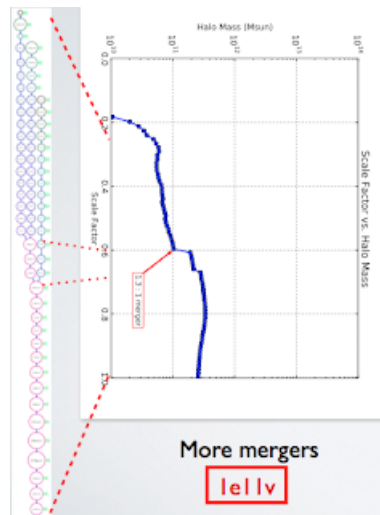
– Density-weighted z-projection of DM density in a 6 Mpc/h box, now with finest resolution particles only

(black box: finest level at the start, white box: contains only the finest particles at the end of simulation, black circle: maximum extent of the particles of the target halo)



– Accretion history: main branch evolution of M_{200}

(M_{200} defined as the mass enclosed in a sphere of mean density contrast of 200 w.r.t. the critical density, measured in the low-resolution simulation)



– Movies of DM-only zoom-in simulation: z-projection of DM density in a 9 Mpc/h box

<http://www.jihoonkim.org/agora/z-movie-1e11v.mov>

– MUSIC parameter file (note it is for DM-only sims)

[ics_60Mpc_1e11v_nocv_enzo.conf](#)

5. Chosen Halos: $\sim 1e12 M_{\text{sun}}$ Halos

[5-1] Quiescent Halo

– $M_{\text{vir}} \sim 1.7 \times 10^{12} M_{\text{sun}}$ @ $z=0$

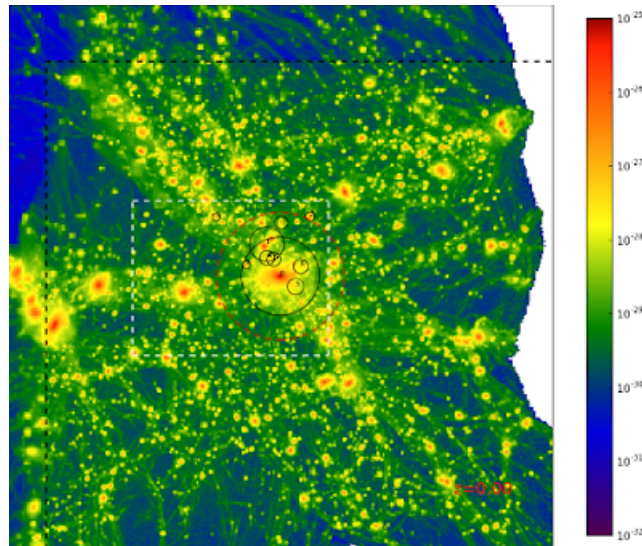
(M_{vir} defined as the mass enclosed in a sphere of mean density contrast of 360 w.r.t. the background density, measured in the low-resolution simulation)

– Lagrangian region of a sphere of radius 998 kpc @ $z=0$

+ a buffer to prohibit contaminations by massive particles inside the halo @ $z=0$ ($512 \times 576 \times 448$)

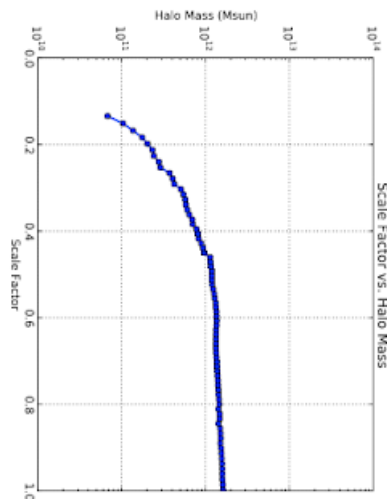
– Density-weighted z-projection of DM density in a 6 Mpc/h box, now with finest resolution particles only

(black box: finest level at the start, white box: contains only the finest particles at the end of simulation, black circle: maximum extent of the particles of the target halo)



– Accretion history: main branch evolution of M_{200}

(M_{200} defined as the mass enclosed in a sphere of mean density contrast of 200 w.r.t. the critical density, measured in the low-resolution simulation)



– Movies of DM-only zoom-in simulation: z-projection of DM density in a 9 Mpc/h box

<http://www.jihoonkim.org/agora/z-movie-1e12q.mov>

– MUSIC parameter file (note it is for DM-only sims)

(1) [ics_60Mpc_1e12q_nocv_enzo.conf](#)

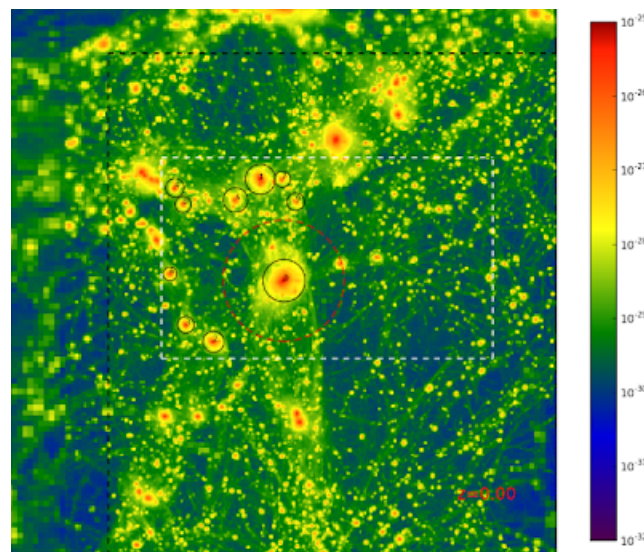
(NOTE: this is an old parameter with a cuboidal highest resolution region by Ji-hoon Kim; not to be used any longer, still here only for an archival purpose; Nov. 20, 2012)

(2) [Buehlmann&Hahn_1e12q_lv12.conf](#)

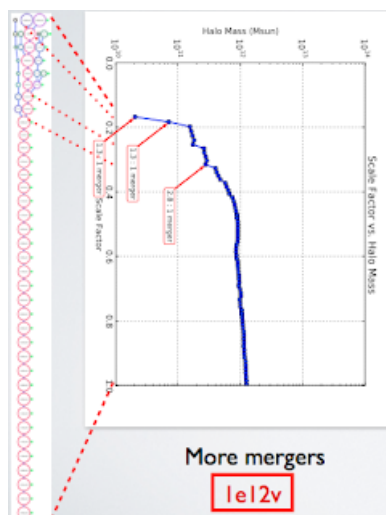
(NOTE: this is a newly selected parameter for **Paper "CGM"** by Michael Buehlmann and Oliver Hahn, of the same halo as above but with an ellipsoidal highest resolution region with a Lagrangian region of $4R_{\text{vir}}$; **z=0 snapshot of a DM-only test with levelmax=11**; **movie**; change levelmax to adjust resolution; Jul. 12, 2017)

[5-2] Violent Halo

- $M_{\text{vir}} \sim 1.3 \times 10^{12} M_{\text{sun}} @z=0$
(M_{vir} defined as the mass enclosed in a sphere of mean density contrast of 360 w.r.t. the background density, measured in the low-resolution simulation)
- Lagrangian region of a sphere of radius 946 kpc @z=0
+ a buffer to prohibit contaminations by massive particles inside the halo @z=0 ($384 \times 384 \times 512$)
- Density-weighted z-projection of DM density in a 6 Mpc/h box, now with finest resolution particles only
(black box: finest level at the start, white box: contains only the finest particles at the end of simulation, black circle: maximum extent of the particles of the target halo)



- Accretion history: main branch evolution of M_{200}
(M_{200} defined as the mass enclosed in a sphere of mean density contrast of 200 w.r.t. the critical density, measured in the low-resolution simulation)



– Movies of DM-only zoom-in simulation: z-projection of DM density in a 9 Mpc/h box

<http://www.jihoonkim.org/agora/z-movie-1e12v.mov>

– MUSIC parameter file (note it is for DM-only sims)

(1) [ics_60Mpc_1e12v_nocv_enzo.conf](#)

(NOTE: this is a parameter set with a cuboidal highest resolution region by Ji-hoon Kim; Nov. 20, 2012)

(2) A newer parameter set may be available if requested

(NOTE: this is a newly selected parameter for the **FIRE Collaboration** by Andrew Wetzel et al., of the same halo as above but with an ellipsoidal highest resolution region with a Lagrangian region of $5R_{\text{vir}}$; please talk to Ji-hoon Kim or Andrew Wetzel; Jan. 20, 2019)

6. Chosen Halos: $\sim 1e13$ Ms Halos

[6-1] Quiescent Halo

– $M_{\text{vir}} \sim 7.5 \times 10^{12} M_{\text{sun}}$ @z=0, $M_{\text{vir}} \sim 1.2 \times 10^{12} M_{\text{sun}}$ @z=2

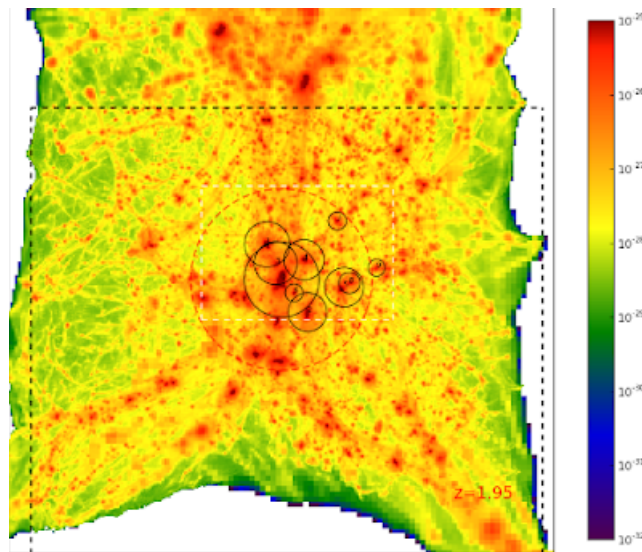
(M_{vir} at z=0 defined as the mass enclosed in a sphere of mean density contrast of "360" w.r.t. the background density, whereas M_{vir} at z=2 defined as the mass enclosed in a sphere of mean density contrast of "200" w.r.t. the background density, measured in the low-resolution simulation)

– Lagrangian region of a sphere of radius 481 kpc @z=2

+ a buffer to prohibit contaminations by massive particles inside the halo @z=2 ($448 \times 512 \times 512$)

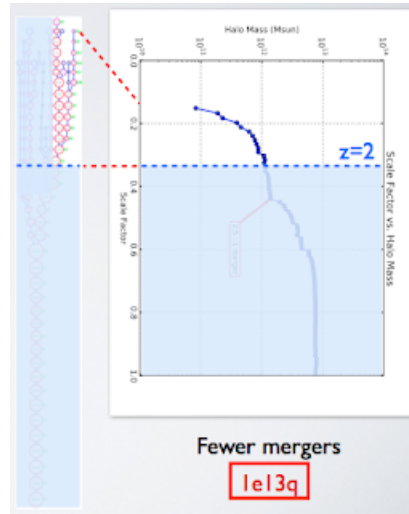
– Density-weighted z-projection of DM density in a 6 Mpc/h box, now with finest resolution particles only

(black box: finest level at the start, white box: contains only the finest particles at the end of simulation, black circle: maximum extent of the particles of the target halo)



– Accretion history: main branch evolution of M_{200} , and the merger tree focusing only down to z=2

(M_{200} defined as the mass enclosed in a sphere of mean density contrast of 200 w.r.t. the critical density, measured in the low-resolution simulation) (alternatively, see the top panel of Figure 1 in [Fiacconi et al. 2016](#) – PonoSQ)



– Movies of DM-only zoom-in simulation: z-projection of DM density in a 9 Mpc/h box down to $z=2$

<http://www.jihoonkim.org/agora/z-movie-1e13q.mov>

– MUSIC parameter file (note it is for DM-only sims)

(1) [ics_60Mpc_1e13q_nocv_enzo.conf](#)

(NOTE: this is an old parameter with a cuboidal highest resolution region by Ji-hoon Kim; not to be used any longer, still here only for an archival purpose; Nov. 20, 2012)

(2) [Buehlmann&Hahn_1e13q_lv12_z2.conf](#)

(NOTE: this is a newly selected parameter for Paper "Clumps" by Michael Buehlmann and Oliver Hahn, of the same halo as above but with an ellipsoidal highest resolution region with a Lagrangian region of $4R_{\text{vir}}$; **$z=2$ snapshot of a DM-only test with levelmax=11; comparison of IC regions**; change levelmax to adjust resolution; Jul. 17, 2017)

[6-2] Violent Halo

– $M_{\text{vir}} \sim 1.4 \times 10^{13} M_{\text{sun}} @z=0$, $M_{\text{vir}} \sim 3.1 \times 10^{12} M_{\text{sun}} @z=2$

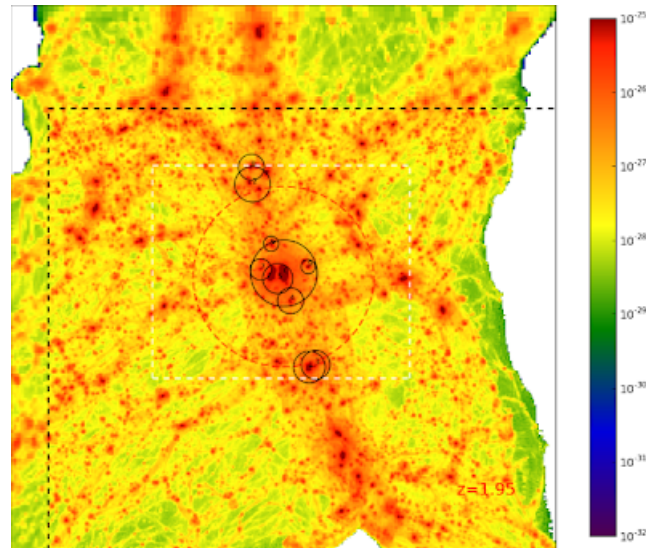
(M_{vir} at $z=0$ defined as the mass enclosed in a sphere of mean density contrast of "360" w.r.t. the background density, whereas M_{vir} at $z=2$ defined as the mass enclosed in a sphere of mean density contrast of "200" w.r.t. the background density, measured in the low-resolution simulation)

– Lagrangian region of a sphere of radius 479 kpc @ $z=2$

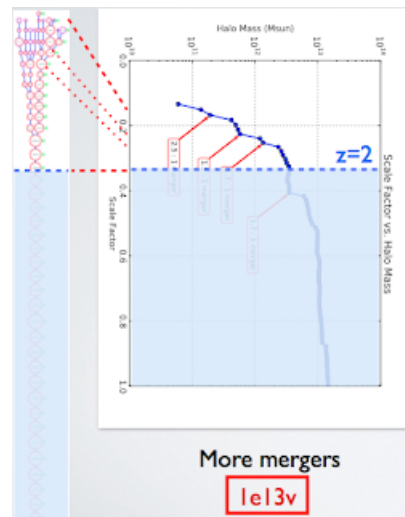
+ a buffer to prohibit contaminations by massive particles inside the halo @ $z=2$ ($512 \times 448 \times 512$)

– Density-weighted z-projection of DM density in a 6 Mpc/h box, now with finest resolution particles only

(black box: finest level at the start, white box: contains only the finest particles at the end of simulation, black circle: maximum extent of the particles of the target halo)



- Accretion history: main branch evolution of M_{200} , and the merger tree focusing only down to $z=2$ (M_{200} defined as the mass enclosed in a sphere of mean density contrast of 200 w.r.t. the critical density, measured in the low-resolution simulation) (alternatively, see the bottom panel of Figure 1 of [Fiacconi et al. 2016](#) – PonoV)



- Movies of DM-only zoom-in simulation: z-projection of DM density in a 9 Mpc/h box down to $z=2$
<http://www.jihoonkim.org/agora/z-movie-1e13v.mov>
- MUSIC parameter file (note it is for DM-only sims)
[ics_60Mpc_1e13v_nocv_enzo.conf](#)

7. Comparing Runs with Different Cosmological Parameters

[7-1] Adopted Cosmology

– We compare two Enzo simulations using two ICs for a $\sim 1e11$ Msun halo, one based on WMAP7 cosmology and the other on Planck cosmology. We adopt runtime parameters of RUN-1 described in [this page](#).

– WMAP7+BAO+H0 (Komatsu et al. 2011; adopted in all of the ICs above)
 $h=0.702$, $\Omega_M=0.272$, $\Omega_{\Lambda}=0.728$, $\Omega_b=0.0455$, $n_s=0.961$ and $\sigma_8=0.807$

– Planck+WP+highL+BAO (Ade et al. 2013; see [this table](#) or Fig.5 of [this paper](#) for comparison of parameters)
 $h=0.678$, $\Omega_M=0.308$, $\Omega_{\Lambda}=0.692$, $\Omega_b=0.0482$, $n_s=0.961$ and $\sigma_8=0.826$

[7-2] (Preliminary) Results

	WMAP7 cosmology	Planck cosmology ^[1]
Density-weighted projection of dark matter density ^[2]	3Mpc/h	3Mpc/h-Planck
Distribution of finest dark matter particles	3Mpc/h-2	3Mpc/h-Planck-2

[note 1] Results here are very preliminary at this stage and subject to change a lot.

[note 2] Projected along the z-axis at $z=0$ in a 3 Mpc/h box centered on the target halo, with only the finest resolution particles in a slab of 3 Mpc/h depth, just as the images in [this page](#)

Working Group III - Blog

[Some preliminary ICs posted](#) We have posted preliminary ICs and MUSIC parameters for the target halos of masses $1e11 - 1e13$ Msun with quiescent merger histories. More will be posted as the progress is made.

Posted Oct 5, 2012, 9:30 AM by Ji-hoon Kim

[First Blog Post](#) This is the first blog post for the Working Group III – Common ICs: Cosmological. Romain and Oliver ran a RAMSES DM-only simulation of the $1e11$ Msol halo. We used a mask to define the high resolution region. See image and profile attached.

Posted Dec 10, 2012, 9:39 PM by Romain Teyssier

Showing posts 1 – 2 of 2. [View more »](#)

